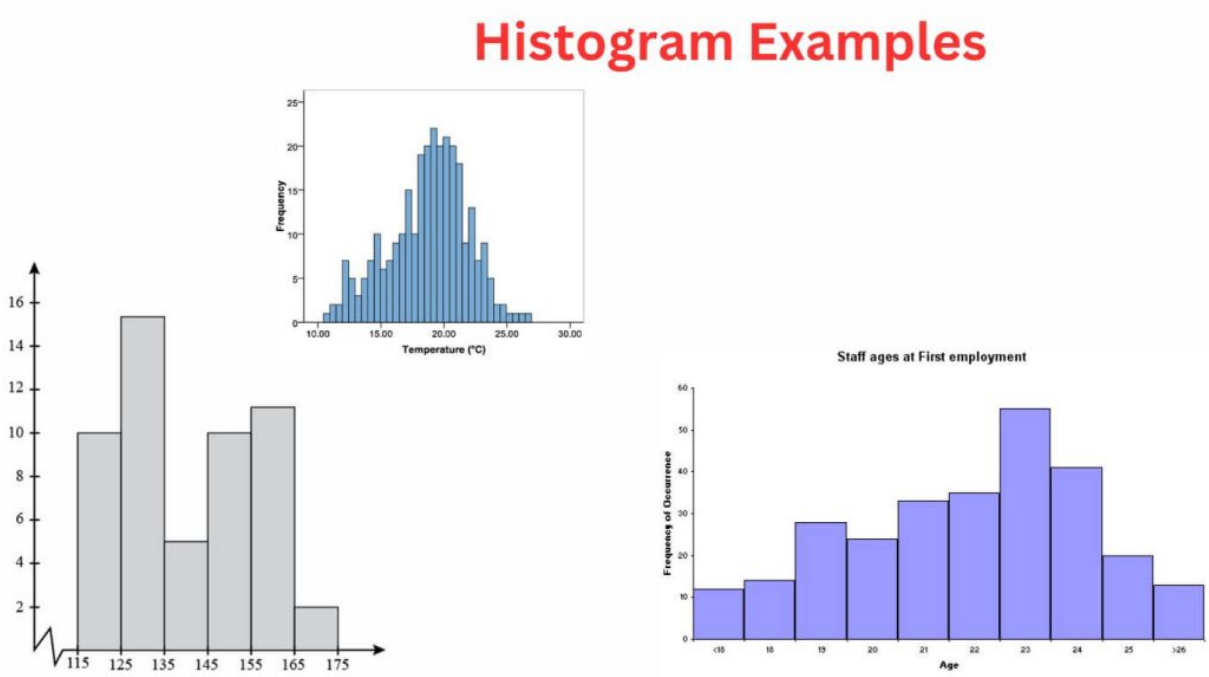
Histogram – Skewness - Kurtosis

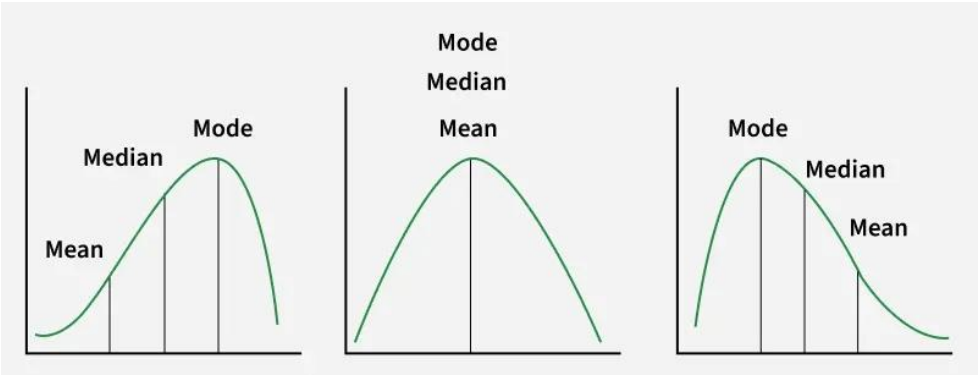
Histogram

A **histogram** is a graphical representation of data distribution. It divides the data into intervals, called bins, and displays the frequency of data points in each bin. Histograms help identify patterns such as symmetry, skewness, modality (number of peaks), and the presence of outliers. They provide a visual summary of the data, making it easier to understand the overall shape and spread before applying numerical measures like skewness or kurtosis.



Skewness

Skewness quantifies the asymmetry of a distribution around its mean. It indicates whether data points are more spread on one side:



Negative Skew **Zero Skew** **Positive Skew**

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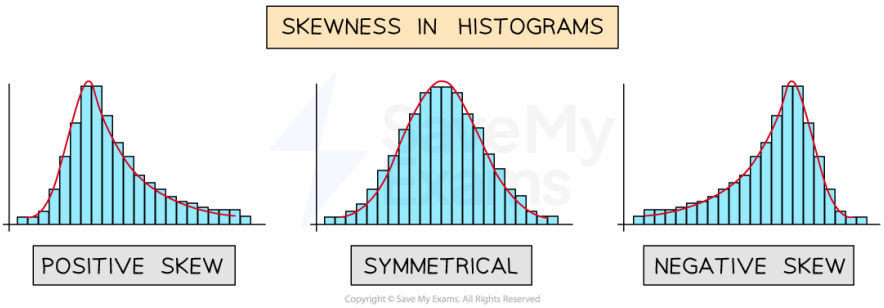

- **Positive Skew (Right-Skewed)**: Tail extends to the right; **Mean > Median > Mode**. Common in income distributions.
- **Negative Skew (Left-Skewed)**: Tail extends to the left; **Mean < Median < Mode**. Seen in easy test scores.
- **Zero Skew (Symmetric)**: Mean, median, and mode are equal; data is evenly distributed.

Visual Detection in Histograms

- **Right tail longer** → Positive skew.
- **Left tail longer** → Negative skew.
- **Equal tails** → No skewness.

Kurtosis

Kurtosis measures the "tailedness" or peak sharpness of a distribution, which can be visualized in a histogram as flat, normal, or sharply peaked shapes.

- **Mesokurtic (kurtosis ≈ 3, excess kurtosis = 0)**: Normal distribution; moderate tails and peak.
- **Leptokurtic (high kurtosis)**: Sharp peak, heavy tails;
  higher probability of extreme values.
- **Platykurtic (low kurtosis):** Flat peak, light tails;
  lower probability of extreme values.

- **Excess kurtosis = 0**: mesokurtic (normal-like)
- **Excess kurtosis < 0**: platykurtic (flatter than normal)
- **Excess kurtosis > 0**: leptokurtic (sharper than normal)

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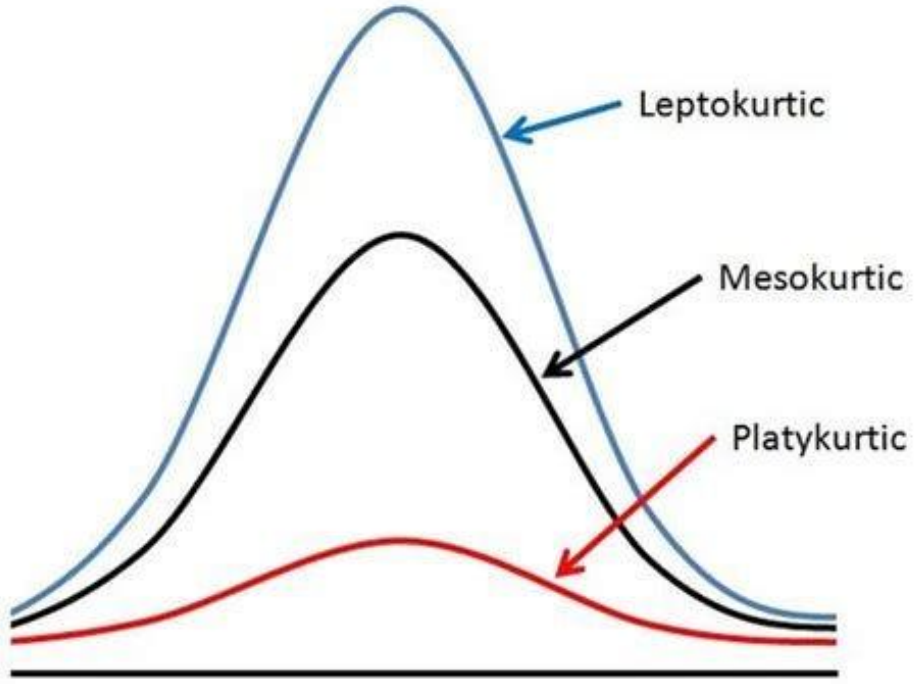

descriptive=pd.DataFrame(index=["Mean","Median","Mode","Q1-25%","Q2-50%","Q3-75%","99%","Q4-100%","IQR","1.5rule","Lesser","Greater","Min","Max","kurtosis","skew"],columns=quan)

```
descriptive[columnName]["kurtosis"]=dataset[columnName].kurtosis()
descriptive[columnName]["skew"]=dataset[columnName].skew()
```

Interpreting Together

They help in,

- Testing for normality
- Detecting outliers
- Informing data transformations
- Supporting statistical modeling and risk assessment.

Histograms visualize data distribution, skewness measures asymmetry, and kurt osis measures the peakedness or tail extremity of a dataset.

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